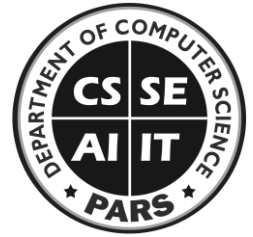




UNIVERSITY OF AGRICULTURE

Department of Computer Science



Lab 6: Programming Fundamentals

(CS-308)

Basic Information			
Lab #:	6	Teacher:	Arsal Mahmood
Objectives	1. Algorithms, flowcharts, pseudocodes		
Registration#	2025-ag-11397	Student Name	Muhammad Arsal
Total Marks	10	Marks Obtained	
Tools	Dev C++ / Visual Studio		

Activity 1 : Algorithms, Flowcharts, Pseudocodes

Understand the questions given below and write Pseudocode, Algorithm and draw flowchart of the problems given below followed by code snippet and output.

1. Write a program that calculates the electricity bill according to the following conditions:

- First 100 units → Rs. 5 per unit
- Next 200 units → Rs. 7 per unit
- Remaining units → Rs. 10 per unit
- The program should input the number of units consumed and display the total bill.

```
#include <iostream>
using namespace std;
int main() {
    double units, total_bill = 0;
    cout << "Enter the number of units consumed: ";
    cin >> units;    if (units <= 100) {        total_bill =
units * 5;    } else if (units <= 300) {
        total_bill = (100 * 5) + ((units - 100) * 7);
    } else {
        total_bill = (100 * 5) + (200 * 7) + ((units - 300) * 10);
    }
}
```

```

    }
    cout << "Total electricity bill: Rs. " << total_bill << endl;
return 0;
}

```

```

Enter the number of units consumed: 234
Total electricity bill: Rs. 1438

-----
Process exited after 3.061 seconds with return value 0
Press any key to continue . . . █

```

2. Calculate the sum of numbers from 1 to 100 and display it on screen by using for loop.

```

#include <iostream> using
namespace std; int main(){
int num1,num2,sum;
cout<<"Enter First Number:";
cin>>num1; cout<<"Enter
second Number:"; cin>>num2;
for(int i=num1;i<=num2;i++){
    sum=sum+i;
}

    cout<<"The Sum is "<<sum;

    return 0;

}

```

```

Enter First Number:1
Enter second Number:100
The Sum is 5050

-----
Process exited after 3.531 seconds with return value 0
Press any key to continue . . . █

```

3. Calculate sum of even and odd numbers separately from a user given range by using for loop. Like if user gives lower_limit of 50 and higher_limit of 80. Calculate sum of all odds and evens in between 50 and 80 and display on screen.

```

#include <iostream>
using namespace std;
int main() {
    int lower_limit, higher_limit;
    int sum_even = 0, sum_odd = 0;
    cout << "Enter lower limit: ";
    cin >> lower_limit;

```

```

    cout << "Enter higher limit: ";
    cin >> higher_limit;
    for (int i = lower_limit; i <= higher_limit; i++) {
        if (i % 2 == 0) {
            sum_even += i;
        } else {
            sum_odd += i;
        }
    }
    cout << "Sum of even numbers: " << sum_even << endl;
    cout << "Sum of odd numbers: " << sum_odd << endl;
    return 0;
}

```

```

Enter lower limit: 10
Enter higher limit: 50
Sum of even numbers: 630
Sum of odd numbers: 600

-----
Process exited after 8.679 seconds with return value 0
Press any key to continue . . .

```

4. Write a program that inputs three numbers and determines the largest number using selection statements.

```

#include <iostream> using namespace std; int main() {
    double num1, num2, num3;    cout << "Enter three
    numbers: ";    cin >> num1 >> num2 >> num3;    if
    (num1 >= num2 && num1 >= num3) {        cout <<
    "The largest number is: " << num1 << endl;    } else if
    (num2 >= num1 && num2 >= num3) {        cout <<
    "The largest number is: " << num2 << endl;
    } else {        cout << "The largest number is: " <<
    num3 << endl;
    }
    return 0;
}

```

```

Enter three numbers: 5 6 7
The largest number is: 7

-----
Process exited after 2.414 seconds with return value 0
Press any key to continue . . .

```

5. Input the marks of a class of ten students. The marks must be between **0** to **100**. Also check marks should not be less than 0 or greater than 100. Calculate the grades of every student and display it by using for loop. Also calculate the total of the grades and the class average.

P-T-O

- $>90 = A$
- $<90 \text{ and } >80 = B$
- $<80 \text{ and } >70 = C$
- $<70 \text{ and } >60 = D$
- $<60 = F$

```
#include <iostream> using namespace std; int main() {    float
marks[10];    float sum = 0;    for (int i = 0; i < 10; i++) {
cout << "Enter marks for student " << i + 1 << " (0-100): ";
cin >> marks[i];    while (marks[i] < 0 || marks[i] > 100) {
cout << "Invalid input. Enter marks between 0 and 100: ";
cin >> marks[i];
    }
    sum += marks[i];
cout << "Grade: ";    if
(marks[i] > 90) {        cout
<< "A" << endl;    } else if
(marks[i] > 80) {        cout
<< "B" << endl;    } else if
(marks[i] > 70) {        cout
<< "C" << endl;    } else if
(marks[i] > 60) {        cout
<< "D" << endl;
    } else {
        cout << "F" << endl;
    }
}
}

float average = sum / 10;    cout << "\nTotal
Marks: " << sum << endl;    cout << "Class
Average: " << average << endl;    return 0;
}
```

```

Enter marks for student 1 (0-100): 90
Grade: B
Enter marks for student 2 (0-100): 98
Grade: A
Enter marks for student 3 (0-100): 89
Grade: B
Enter marks for student 4 (0-100): 70
Grade: D
Enter marks for student 5 (0-100): 78
Grade: C
Enter marks for student 6 (0-100): 89
Grade: B
Enter marks for student 7 (0-100): 90
Grade: B
Enter marks for student 8 (0-100): 78
Grade: C
Enter marks for student 9 (0-100): 78
Grade: C
Enter marks for student 10 (0-100): 98
Grade: A

Total Marks: 858
Class Average: 85.8

-----
Process exited after 18.89 seconds with return value 0
Press any key to continue . . .

```

6. Calculate sum of even and off numbers by using do-while loop separately from a user given range by using for loop. Like if user gives **lower_limit** of 50 and **higher_limit** of 80. Calculate sum of all odds and even in between 50 and 80 and display on screen.

```

#include <iostream>
using namespace std;
int main() {
    int lower_limit, higher_limit;
    int sum_even = 0, sum_odd = 0;
    cout << "Enter lower limit: ";
    cin >> lower_limit;    cout <<
    "Enter higher limit: ";    cin >>
    higher_limit;    int i =
    lower_limit;    if (i <=
    higher_limit) {        do {

```

```

        if (i % 2 == 0) {
sum_even += i;
        } else {
sum_odd += i;
        }
i++;
    } while (i <= higher_limit);
}
cout << "Sum of even numbers: " << sum_even << endl;
cout << "Sum of odd numbers: " << sum_odd << endl;
return 0;
}

```

```

Enter lower limit: 10
Enter higher limit: 100
Sum of even numbers: 2530
Sum of odd numbers: 2475

-----
Process exited after 9.517 seconds with return value 0
Press any key to continue . . .

```

7. Print the table of user given number from 1 upto user given limit by using for loop. Like if user enters 3, and limit is 4, program should print table of 3 as follows:

3	*	1	=	03
3	*	2	=	06
3	*	3	=	09
3	*	4	=	12

```

#include <iostream> using namespace std;
int main() {    int num, limit;    cout <<
"Enter the number for the table: ";    cin >>
num;    cout << "Enter the limit: ";    cin >>
limit;    for (int i = 1; i <= limit; i++) {
int result = num * i;    cout << num << "
* " << i << " = ";    if (result < 10) {
cout << "0";
    }
    cout << result << endl;
    }
return 0;
}

```

```

Enter the number for the table: 3
Enter the limit: 5
3 * 1 = 03
3 * 2 = 06
3 * 3 = 09
3 * 4 = 12
3 * 5 = 15

-----
Process exited after 6.862 seconds with return value 0
Press any key to continue . . .

```

8. Write a program to find the factorial of a number using a for loop.

```

#include <iostream> using
namespace std; int main() {
int num; long long factorial
= 1; cout << "Enter a
number: "; cin >> num;

if (num < 0) { cout << "Factorial of a negative number
doesn't exist." << endl;
} else { for (int i = 1; i <=
num; i++) {
factorial *= i;
}
cout << "Factorial of " << num << " = " << factorial << endl;
}
return 0;
}

```

```

Enter a number: 5
Factorial of 5 = 120

-----
Process exited after 2.219 seconds with return value 0
Press any key to continue . . .

```

9. Write a program that takes an integer and prints its reverse using a for loop.

```

#include <iostream> using namespace std; int
main() { int num, reversedNum = 0; cout
<< "Enter an integer: "; cin >> num; for (;
num != 0; num /= 10) { int digit = num %

```

```

10;    reversedNum = reversedNum * 10 +
digit;
}
cout << "Reversed number: " << reversedNum << endl;
return 0;
}

```

```

Enter an integer: 14
Reversed number: 41

-----
Process exited after 3.707 seconds with return value 0
Press any key to continue . . .

```

10. Write a program to check whether a number is a palindrome (same forward and backward) using a for loop.

```

#include <iostream>
using namespace std;
int main() {
    int num, originalNum, reversedNum = 0;
    cout << "Enter an integer: ";    cin >>
num;    originalNum = num;    for (; num
> 0; num /= 10) {        int digit = num %
10;
        reversedNum = reversedNum * 10 + digit;
    }
    if (originalNum == reversedNum) {
        cout << originalNum << " is a palindrome." << endl;
    } else {
        cout << originalNum << " is not a palindrome." << endl;
    }
    return 0;
}

```

```

Enter an integer: 9
9 is a palindrome.

-----
Process exited after 3.83 seconds with return value 0
Press any key to continue . . .

```

11. Write a program to calculate x^y (x raised to the power y) using a for loop.


```

#include <iostream>
using namespace std;
int main() {
    int x, y;
    long long result = 1;    cout <<
    "Enter base (x): ";    cin >> x;
    cout << "Enter exponent (y): ";
    cin >> y;    if (y < 0) {
        cout << "Exponent should be a non-negative integer." << endl;
    } else {        for (int i = 1; i
    <= y; i++) {            result *= x;

        }
        cout << x << "^" << y << " = " << result << endl;
    }
    return 0;
}

```

```

Enter base (x): 5
Enter exponent (y): 6
5^6 = 15625

-----
Process exited after 16.65 seconds with return value 0
Press any key to continue . . .

```

12. Print a diamond shape on screen by using for loop.

```

*

*   *   *

*   *   *   *   *

*   *   *   *   *   *   *

*   *   *   *   *

*   *   *

*

```

```

#include <iostream> using
namespace std; int main() {    int
n = 4;    for (int i = 1; i <= n; i++)
{        for (int j = 1; j <= n - i;
j++) {            cout << " ";

        }

        for (int k = 1; k <= (2 * i - 1); k++) {
            cout << "* ";

```

```

    }
    cout << endl;
}
for (int i = n - 1; i >= 1; i--) {
for (int j = 1; j <= n - i; j++) {
cout << " ";
    }
    for (int k = 1; k <= (2 * i - 1); k++) {
        cout << "* ";
    }
    cout << endl;
}
return 0;
} #include <iostream> using
namespace std; int main() {    int
n = 4;    for (int i = 1; i <= n; i++)
{        for (int j = 1; j <= n - i;
j++) {            cout << " ";
        }
        for (int k = 1; k <= (2 * i - 1); k++) {
cout << "* ";
        }
        cout << endl;
    }
    for (int i = n - 1; i >= 1; i--) {
for (int j = 1; j <= n - i; j++) {
cout << " ";
    }
    for (int k = 1; k <= (2 * i - 1); k++) {
cout << "* ";
    }
    cout << endl;
}
return 0;
}

```

```
      *
    * * *
  * * * * *
* * * * * * *
  * * * * *
    * * *
      *

-----
Process exited after 0.01783 seconds with return value 0
Press any key to continue . . .
```

13. Write a program that inputs a number and prints the **reverse of the number** using a loop.

```
#include <iostream> using namespace std; int
main() { int num, reversedNum = 0; cout
<< "Enter a number: "; cin >> num; while
(num != 0) { int digit = num % 10;
reversedNum = reversedNum * 10 + digit;
num /= 10;
}
cout << "Reversed number: " << reversedNum << endl;
return 0;
}
```

```
Enter a number: 67
Reversed number: 76

-----
Process exited after 2.771 seconds with return value 0
Press any key to continue . . .
```

{end of Lab}